

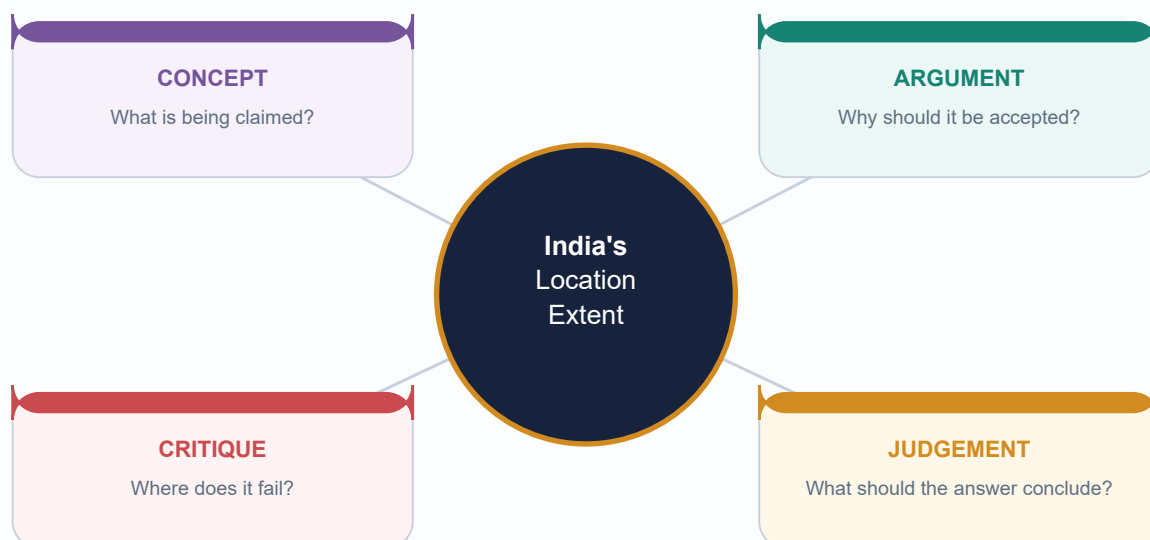
UPSC GEOGRAPHY TOPIC PACKAGE

India's Location & Extent Complete Topic Package

Subject: Geography | Paper: GS-I

Sources: 01_The-Earth-and-the-Universe.md + 01_India-Location-and-Extent.md

Practice is derived from explicit chapter traps, not generic placeholder questions



Facts from official sources | Inferences clearly marked | No fabricated data

HIGH UPSC RELEVANCE

HIGH

Foundation · Core Foundation

GS-I Geography

INTRO & ORIGIN

Subject: Geography · **Tier:** Must-Do (foundation + standard) · **GS Paper:** GS-I

Foundation learning layer: concepts → constitutional detail → UPSC application.

CORE FOUNDATION: RECALL MAP

Core Foundation

V

Dimension 1 Subject: Geography · Tier: Must-Do (foundation + standard) · GS Paper: GS-I	Dimension 2 Grounded in: G.C. Leong, Certificate Physical & Human Geography + Majid Husain, Indian & World Geography.
Dimension 3 □ = from source book · □□ = inference / standard knowledge · □ = current affairs.	Dimension 4 Companion: advanced/01India-Location-and-Extent.md.

STATIC THEORY

- **Grounded in:** G.C. Leong, *Certificate Physical & Human Geography* + Majid Husain, *Indian & World Geography*.

- □ = from source book · □□ = inference / standard knowledge · □ = current affairs.

- *Companion: advanced/01_India-Location-and-Extent.md.*

MUST-KNOW FACTS

1. = from source book · □□ = inference / standard knowledge · □ = current affairs.

MAINS ANGLE

STUDY LINK

India's Location & Extent → Core Foundation

HIGH

Foundation · Shape & Motions of the Earth (G.C. Leong)

GS-I Geography

INTRO & ORIGIN

□ The Earth is approximately an **oblate spheroid**: flattened at the poles and bulging at the equator. A **geoid** is the irregular, gravity-defined mean-sea-level surface, not simply another name for an oblate spheroid. Proofs of broad sphericity include the circular horizon, ships' hulls vanishing first, curved shadow in lunar eclipses, photos from space.

Foundation learning layer: concepts → constitutional detail → UPSC application.

SHAPE & MOTIONS OF THE EARTH (G.C. LEONG): RECALL MAP

Shape & Motions of the Earth (G.C. Leong)

V

Dimension 1 ☐ The Earth is approximately an oblate spheroid: flattened at the poles and bulging at the equator. A geoid is the irregular, gravity-defined mean-sea	Dimension 2 ☐ Two motions drive time and seasons:
Dimension 3 Rotation — spin on axis, ~24 hours (~1,675 kmph at equator) → day & night.	Dimension 4 Revolution — orbit round the Sun, 365.25 days → the year (leap-year adjustment).

STATIC THEORY

- ☐ Two motions drive time and seasons:
- **Rotation** — spin on axis, ~24 hours (~1,675 kmph at equator) → **day & night**.
- **Revolution** — orbit round the Sun, **365.25 days** → **the year** (leap-year adjustment).
- **Axial tilt** = about **23.5° from the perpendicular to the orbital plane** (66.5° to the plane) → **seasons** and changing day length.
- ☐ Trap: **Rotation causes day & night**; the **23.5° tilt + revolution** cause seasons — *not* rotation.

MEMORY HOOK

Trap: Rotation causes day & night; the 23.5° tilt + revolution cause seasons — *not* rotation.

MUST-KNOW FACTS

1. The Earth is approximately an **oblate spheroid**: flattened at the poles and bulging at the equator. A **geoid** is the irregular, gravity-defined mean-sea-level surface, not simply another name for an oblate spheroid. Proofs of broad sphericity include the circular horizon, ships' hulls vanishing first, curved shadow in lunar eclipses, photos from space.
2. Two motions drive time and seasons:

MAINS ANGLE

STUDY LINK

India's Location & Extent → Shape & Motions of the Earth (G.C. Leong)

HIGH

Foundation · Latitude, Longitude & Time

GS-I Geography

INTRO & ORIGIN

Structured comparison from the foundation source layer.

KEY DATA

Concept	Rule	Consequence
☐ Latitude	Angular distance N/S of Equator (0–90°)	Locates place; fixes climate belts
☐ Longitude	Angular distance E/W of Prime Meridian (Greenwich, 0°)	Fixes time ; 15° = 1 hour
☐ Key parallels	Tropics 23.5°, Arctic/Antarctic Circles 66.5°	Solar limits
☐ Date Line	≈ 180° meridian (IDL)	Calendar day changes on crossing

MAINS ANGLE

STUDY LINK

India's Location & Extent → Latitude, Longitude & Time

HIGH

Foundation · Latitude, Longitude & Time

GS-I Geography

INTRO & ORIGIN

☐ Earth turns 360° in 24 h → **15° of longitude = 1 hour**; places **east are ahead** in time. ☐ Standard Time uses a central meridian: **India = 82.5°E** → **IST = UTC+5:30**.

Foundation learning layer: concepts → constitutional detail → UPSC application.

STATIC THEORY

- ☐ Trap: **Longitude** fixes time (15° = 1 h); **latitude** fixes climate/position N–S.

MEMORY HOOK

Trap: Longitude fixes time (15° = 1 h); latitude fixes climate/position N–S.

MUST-KNOW FACTS

1. Earth turns 360° in 24 h → **15° of longitude = 1 hour**; places **east are ahead** in time. Standard Time uses a central meridian: **India = 82.5°E** → **IST = UTC+5:30**.

MAINS ANGLE

STUDY LINK

India's Location & Extent → Latitude, Longitude & Time

HIGH

Foundation · Origin of the Universe & Earth's Interior (Majid Husain)

GS-I Geography

INTRO & ORIGIN

Structured comparison from the foundation source layer.

KEY DATA

Layer	Extent	Composition & state
□ Crust	Surface → Moho (~35 km avg)	Rigid; less than 1% of Earth's volume
□ Mantle	~35 km → 2900 km	Solid rock + magma; density 3.3–5.7
□ Outer core	2900 km → ~5150 km	Liquid nickel-iron (NiFe)
□ Inner core	~5150 km → centre	Solid iron-nickel alloy under immense pressure

MAINS ANGLE

STUDY LINK

India's Location & Extent → Origin of the Universe & Earth's Interior (Majid Husain)

HIGH

Foundation · Origin of the Universe & Earth's Interior (Majid Husain)

GS-I Geography

INTRO & ORIGIN

□ The **Big Bang** marks the expansion of the universe from an extremely hot, dense early state; modern estimates place its age at about **13.8 billion years**. The solar system formed by **gravitational collapse of a cold gas–dust cloud** (mainly H & He) in a Milky Way spiral arm. □ Inner **terrestrial** planets = rock/metal (crystallised hot); outer **Jovian** planets = gas/ice condensates (cold).

Foundation learning layer: concepts → constitutional detail → UPSC application.

STATIC THEORY

- □ **Moho** = crust/mantle boundary (~35 km). □ **Gutenberg discontinuity** (~2900 km) = mantle/outer-core boundary. □ **S-waves** (shear) cannot pass liquids and vanish at 2900 km → **outer core is liquid**; **P-waves** pass through all layers.

MUST-KNOW FACTS

1. The **Big Bang** marks the expansion of the universe from an extremely hot, dense early state; modern estimates place its age at about **13.8 billion years**. The solar system formed by **gravitational collapse of a cold gas–dust cloud** (mainly H & He) in a Milky Way spiral arm. Inner **terrestrial** planets = rock/metal (crystallised hot); outer **Jovian** planets = gas/ice condensates (cold).

2. **Moho** = crust/mantle boundary (~35 km). **Gutenberg discontinuity** (~2900 km) = mantle/outer-core boundary. **S-waves** (shear) cannot pass liquids and vanish at 2900 km → **outer core is liquid**; **P-waves** pass through all layers.

MAINS ANGLE

STUDY LINK

India's Location & Extent → Origin of the Universe & Earth's Interior (Majid Husain)

HIGH

Foundation · Must-Know Facts (Prelims)

GS-I Geography

INTRO & ORIGIN

- Earth's geometric approximation = **oblate spheroid**; geoid = gravity-defined mean-sea-level surface.

Foundation learning layer: concepts → constitutional detail → UPSC application.

MUST-KNOW FACTS (PRELIMS): RECALL MAP

Must-Know Facts (Prelims)

V

Dimension 1 □ Earth's geometric approximation = oblate spheroid; geoid = gravity-defined mean-sea-level surface.	Dimension 2 □ Rotation ≈ 24 h (day-night); Revolution = 365.25 days (year); tilt 23.5° (seasons).
Dimension 3 □ 15° longitude = 1 hour; IST meridian = 82.5°E (UTC+5:30).	Dimension 4 □ International Date Line ≈ 180° meridian.

STATIC THEORY

- □ Rotation ≈ 24 h (day-night); Revolution = 365.25 days (year); tilt 23.5° (seasons).
- □ **15° longitude = 1 hour**; IST meridian = **82.5°E (UTC+5:30)**.
- □ International Date Line ≈ **180°** meridian.
- □ Only the **outer core is liquid** (S-wave proof); inner core is solid.
- □ Moho = crust/mantle; Gutenberg = mantle/outer-core.

MUST-KNOW FACTS

1. Earth's geometric approximation = **oblate spheroid**; geoid = gravity-defined mean-sea-level surface.
2. Rotation ≈ 24 h (day-night); Revolution = 365.25 days (year); tilt 23.5° (seasons).
3. **15° longitude = 1 hour**; IST meridian = **82.5°E (UTC+5:30)**.
4. International Date Line ≈ **180°** meridian.
5. Only the **outer core is liquid** (S-wave proof); inner core is solid.
6. Moho = crust/mantle; Gutenberg = mantle/outer-core.

MAINS ANGLE

STUDY LINK

India's Location & Extent → Must-Know Facts (Prelims)

HIGH

Foundation · UPSC Traps

GS-I Geography

INTRO & ORIGIN

Exam-ready treatment of UPSC Traps.

Foundation learning layer: concepts → constitutional detail → UPSC application.

UPSC TRAPS

WRONG: Earth is a perfect sphere

CORRECT: oblate spheroid/geoid.

WRONG: Rotation causes seasons

CORRECT: rotation causes day/night; tilt + revolution cause seasons.

WRONG: Latitude fixes time

CORRECT: longitude fixes time.

WRONG: The whole core is molten

CORRECT: only the outer core is liquid.

WRONG: Moho separates mantle from core

CORRECT: Moho = crust/mantle; Gutenberg = mantle/outer-core.

WRONG: P-waves cannot travel through liquids

CORRECT: S-waves can't; P-waves travel through solids and liquids.

MAINS ANGLE

STUDY LINK

India's Location & Extent → UPSC Traps

HIGH

Foundation · □ Current link

GS-I Geography

NEWS TRIGGER

□ **Time-zone debate:** India's single **IST (UTC+5:30)** vs periodic demands for a separate **North-East time zone** (sun rises ~2 hours earlier in Arunachal than in Saurashtra) keep the longitude-time concept exam-relevant. □ Seismic-wave (P/S) study of the interior is the same physics used to locate earthquake foci in India's high-seismicity Himalayan belt.

INTRO & ORIGIN

Exam-ready treatment of □ Current link.

Foundation learning layer: concepts → constitutional detail → UPSC application.

MAINS ANGLE

STUDY LINK

India's Location & Extent → □ Current link

HIGH

Foundation · Mains angles

GS-I Geography

INTRO & ORIGIN

Exam-ready treatment of Mains angles.

Foundation learning layer: concepts → constitutional detail → UPSC application.

MAINS ANGLE

Explain how **seismic-wave (P/S) behaviour** maps the Earth's layered, differentiated interior — the foundation for plate tectonics, volcanism and India's seismic hazard. The longitude–time relationship and the case for/against a **separate North-East India time zone**.

STUDY LINK

India's Location & Extent → Mains angles

HIGH

Advanced · Core Foundation

GS-I Geography

INTRO & ORIGIN

Subject: Geography · **Tier:** Advanced (India-applied) · **GS Paper:** GS-I

Advanced learning layer: concepts → constitutional detail → UPSC application.

CORE FOUNDATION: RECALL MAP

Core Foundation

V

Dimension 1 Subject: Geography · Tier: Advanced (India-applied) · GS Paper: GS-I	Dimension 2 Grounded in: D.R. Khullar, India: A Comprehensive Geography + Majid Husain, India geography + verified CA.
Dimension 3 □ = from source book · □□ = inference / standard knowledge · □ = current affairs.	Dimension 4 Companion: basic/01The-Earth-and-the-Universe.md (world-physical basics).

STATIC THEORY

- **Grounded in:** D.R. Khullar, India: A Comprehensive Geography + Majid Husain, India geography + verified CA.

- □ = from source book · □□ = inference / standard knowledge · □ = current affairs.

- *Companion: basic/01_The-Earth-and-the-Universe.md (world-physical basics).*

MUST-KNOW FACTS

1. = from source book · □□ = inference / standard knowledge · □ = current affairs.

MAINS ANGLE

STUDY LINK

India's Location & Extent → Core Foundation

HIGH

Advanced · Snapshot

GS-I Geography

INTRO & ORIGIN

Structured comparison from the advanced source layer.

KEY DATA

Parameter	Value	Significance
□ Latitudinal extent	8°4'N – 37°6'N	Tropic of Cancer bisects India; tropical S vs sub-tropical N
□ Longitudinal extent	68°7'E – 97°25'E	~29° → ~2-hr sun-time gap E–W
□ IST meridian	82°30'E (via Mirzapur, UP)	One standard time for the whole country
□ Area / coastline	~3.28 M sq km; 11,098.81 km coastline (official 2025 GIS/HWL remeasurement)	7th-largest country; revised coastal-risk baseline
□ N–S / E–W stretch	~3,214 km / ~2,933 km	Mainland extent

MAINS ANGLE

STUDY LINK
India's Location & Extent → Snapshot

HIGH

Advanced · Location

GS-I Geography

INTRO & ORIGIN

□ India lies **wholly in the Northern & Eastern hemispheres**, between about **8°4'N–37°6'N** and **68°7'E–97°25'E**. □ The **Tropic of Cancer (23.5°N)** passes through the middle of India (**8 states**, Gujarat to Mizoram), dividing it into a **tropical south** and a **sub-tropical north** — a key control on India's climate.

Advanced learning layer: concepts → constitutional detail → UPSC application.

MUST-KNOW FACTS

1. India lies **wholly in the Northern & Eastern hemispheres**, between about **8°4'N–37°6'N** and **68°7'E–97°25'E**. The **Tropic of Cancer (23.5°N)** passes through the middle of India (**8 states**, Gujarat to Mizoram), dividing it into a **tropical south** and a **sub-tropical north** — a key control on India's climate.

MAINS ANGLE

STUDY LINK
India's Location & Extent → Location

HIGH

Advanced · Longitude & Standard Time

GS-I Geography

INTRO & ORIGIN

□ The ~29° longitudinal spread means a **~2-hour difference** in local sun-time between the eastern (Arunachal) and western (Saurashtra/Gujarat) extremes. □ Resolved by adopting a **single Indian Standard Time (UTC+5:30)** at **82°30'E** for administrative uniformity.

Advanced learning layer: concepts → constitutional detail → UPSC application.

LONGITUDE & STANDARD TIME: RECALL MAP

Longitude & Standard Time

V

Dimension 1

- The ~29° longitudinal spread means a ~2-hour difference in local sun-time between the eastern (Arunachal) and western (Saurashtra/Gujarat) extremes.

Dimension 2

- Trap: The Tropic of Cancer (not the Equator) passes through India; India has one time zone,

Dimension 3
not multiple.

STATIC THEORY

- □ Trap: The **Tropic of Cancer** (not the Equator) passes through India; India has **one** time zone,
- not multiple.

MEMORY HOOK

Trap: The Tropic of Cancer (not the Equator) passes through India; India has one time zone,

MUST-KNOW FACTS

1. The ~29° longitudinal spread means a ~2-hour difference in local sun-time between the eastern (Arunachal) and western (Saurashtra/Gujarat) extremes. Resolved by adopting a **single Indian Standard Time (UTC+5:30)** at **82°30'E** for administrative uniformity.

MAINS ANGLE

STUDY LINK

India's Location & Extent → Longitude & Standard Time

HIGH

Advanced · Map hooks

GS-I Geography

INTRO & ORIGIN

Structured comparison from the advanced source layer.

KEY DATA

Line	West-to-east Indian states
□ Tropic of Cancer	Gujarat → Rajasthan → Madhya Pradesh → Chhattisgarh → Jharkhand → West Bengal → Tripura → Mizoram
□ Standard Meridian (82°30'E)	Uttar Pradesh → Madhya Pradesh → Chhattisgarh → Odisha → Andhra Pradesh

MAINS ANGLE

STUDY LINK

India's Location & Extent → Map hooks

HIGH

Advanced · Map hooks

GS-I Geography

INTRO & ORIGIN

❑ Trap: The Tropic of Cancer crosses eight states; the Standard Meridian crosses a different

Advanced learning layer: concepts → constitutional detail → UPSC application.

STATIC THEORY

- north-south set and is conventionally associated with Mirzapur, Uttar Pradesh.

MEMORY HOOK

Trap: The Tropic of Cancer crosses eight states; the Standard Meridian crosses a different

MAINS ANGLE

STUDY LINK

India's Location & Extent → Map hooks

HIGH

Advanced · Must-Know Facts (Prelims)

GS-I Geography

INTRO & ORIGIN

❑ India: 8°4'N–37°6'N and 68°7'E–97°25'E.

Advanced learning layer: concepts → constitutional detail → UPSC application.

MUST-KNOW FACTS (PRELIMS): RECALL MAP

Must-Know Facts (Prelims)

V

Dimension 1 ❑ India: 8°4'N–37°6'N and 68°7'E–97°25'E.	Dimension 2 ❑ Tropic of Cancer (23.5°N) passes through 8 Indian states.
Dimension 3 ❑ IST meridian = 82°30'E (near Mirzapur); UTC+5:30.	Dimension 4 ❑ ~2-hour sun-time gap between Arunachal and Gujarat, yet a single IST.

STATIC THEORY

- ❑ Tropic of Cancer (23.5°N) passes through 8 Indian states.

- ❑ IST meridian = 82°30'E (near Mirzapur); UTC+5:30.

- ❑ ~2-hour sun-time gap between Arunachal and Gujarat, yet a single IST.

- ❑ India's officially remeasured coastline is 11,098.81 km. The legacy

- 7,516.6 km figure used an older measurement method.

MUST-KNOW FACTS

1. India: **8°4'N–37°6'N** and **68°7'E–97°25'E**.
2. **Tropic of Cancer (23.5°N)** passes through **8 Indian states**.
3. IST meridian = **82°30'E** (near Mirzapur); **UTC+5:30**.
4. ~2-hour sun-time gap between Arunachal and Gujarat, yet a single IST.
5. India's officially remeasured coastline is **11,098.81 km**. The legacy

MAINS ANGLE**STUDY LINK**

India's Location & Extent → Must-Know Facts (Prelims)

HIGH**Advanced · UPSC Traps**

GS-I Geography

INTRO & ORIGIN

Exam-ready treatment of UPSC Traps.

Advanced learning layer: concepts → constitutional detail → UPSC application.

UPSC TRAPS

WRONG: The Equator passes through India

CORRECT: it is the **Tropic of Cancer**.

WRONG: India has multiple official time zones

CORRECT: **single IST** (82°30'E, UTC+5:30).

MAINS ANGLE**STUDY LINK**

India's Location & Extent → UPSC Traps

HIGH**Advanced · ☐ Current link**

GS-I Geography

NEWS TRIGGER

- ☐ The recurring "**single time zone vs North-East time zone**" debate applies the longitude-time concept to India — proponents cite energy savings and better work-hours; opponents cite administrative and railway-safety complexity.
- ☐ **Coastline revision (2025):** National Hydrographic Office/Survey of India GIS measurement of the high-water line revised the official length to **11,098.81 km**. It updates planning/risk baselines; it does **not** itself redraw CRZ high-tide-line boundaries. PIB source.

INTRO & ORIGIN

Exam-ready treatment of ☐ Current link.

Advanced learning layer: concepts → constitutional detail → UPSC application.

MAINS ANGLE**STUDY LINK**

India's Location & Extent → ☐ Current link

HIGH

Advanced · Mains angles

GS-I Geography

INTRO & ORIGIN

Exam-ready treatment of Mains angles.

Advanced learning layer: concepts → constitutional detail → UPSC application.

MAINS ANGLE

Use India's longitudinal extent to argue **for/against a separate North-East time zone** — energy and productivity gains vs administrative and railway-safety costs. How India's latitudinal spread (tropical to sub-tropical) shapes its climatic diversity.

STUDY LINK

India's Location & Extent → Mains angles

HIGH

Current-Affairs Verification Note

GS-I Geography

INTRO & ORIGIN

A live February-August 2026 scan was performed. No unverified headline is inserted. The source chapter's dated current anchor is retained where it is already verified.

STATIC THEORY

- Static concepts remain the evidentiary base.
- Current affairs are used only when the conceptual linkage is direct.
- No-current-item is preferable to a fabricated or weakly sourced linkage.

MAINS ANGLE**STUDY LINK**

India's Location & Extent -> Current application

HIGH

Topic-Specific MCQ Loop 1

GS-I Geography

INTRO & ORIGIN

Each question is generated from an explicit source-chapter trap and explanation.

STATIC THEORY

- Q1. Which statement correctly resolves this common misconception: "Earth is a perfect sphere"?

- A. **oblate spheroid/geoid**.

- B. Earth is a perfect sphere

- C. Rotation causes seasons

- D. Latitude fixes time

- ANSWER: A - **oblate spheroid/geoid**.

- Q2. Which statement correctly resolves this common misconception: "Rotation causes seasons"?

- A. Rotation causes seasons

- B. rotation causes day/night; **tilt + revolution** cause seasons.

- C. Latitude fixes time

- D. The whole core is molten

- ANSWER: B - rotation causes day/night; **tilt + revolution** cause seasons.

- Q3. Which statement correctly resolves this common misconception: "Latitude fixes time"?

- A. Latitude fixes time

- B. The whole core is molten

- C. **longitude** fixes time.

- D. Moho separates mantle from core

- ANSWER: C - **longitude** fixes time.

- Q4. Which statement correctly resolves this common misconception: "The whole core is molten"?

- A. The whole core is molten

- B. Moho separates mantle from core

- C. P-waves cannot travel through liquids

- D. **only the outer core** is liquid.

- ANSWER: D - **only the outer core** is liquid.

MEMORY HOOK

Resolve the misconception before memorising the fact.

MAINS ANGLE

STUDY LINK

India's Location & Extent -> Source-grounded practice

HIGH

Topic-Specific MCQ Loop 2

GS-I Geography

INTRO & ORIGIN

Each question is generated from an explicit source-chapter trap and explanation.

STATIC THEORY

- Q5. Which statement correctly resolves this common misconception: "Moho separates mantle from core"?
 - A. Moho = **crust/mantle**; Gutenberg = mantle/outer-core.
 - B. Moho separates mantle from core
 - C. P-waves cannot travel through liquids
 - D. The Equator passes through India
- ANSWER: A - Moho = **crust/mantle**; Gutenberg = mantle/outer-core.
- Q6. Which statement correctly resolves this common misconception: "P-waves cannot travel through liquids"?
 - A. P-waves cannot travel through liquids
 - B. **S-waves** can't; **P-waves travel through solids and liquids.**
 - C. The Equator passes through India
 - D. India has multiple official time zones
- ANSWER: B - **S-waves** can't; **P-waves travel through solids and liquids.**
- Q7. Which statement correctly resolves this common misconception: "The Equator passes through India"?
 - A. The Equator passes through India
 - B. India has multiple official time zones
 - C. it is the **Tropic of Cancer.**
 - D. Earth is a perfect sphere
- ANSWER: C - it is the **Tropic of Cancer.**
- Q8. Which statement correctly resolves this common misconception: "India has multiple official time zones"?
 - A. India has multiple official time zones
 - B. Earth is a perfect sphere
 - C. Rotation causes seasons
 - D. **single IST** (82°30'E, UTC+5:30).
- ANSWER: D - **single IST** (82°30'E, UTC+5:30).

MEMORY HOOK

Resolve the misconception before memorising the fact.

MAINS ANGLE**STUDY LINK**

India's Location & Extent -> Source-grounded practice

HIGH**Solved 10-Mark Practice****GS-I Geography****INTRO & ORIGIN**

Explain India's Location & Extent with reference to its concepts, applications and limitations.

ANSWER ARCHITECTURE**1. Concept**

Define the exact scope and terminology.

V**2. Core analysis**

Organise the source content into two or three dimensions.

V**3. Evidence**

Use the chapter's India-centric examples or chronology.

V**4. Critical layer**

Add traps, limitations, counter-view or implementation constraints.

V**5. Judgement**

Conclude in direct response to the directive.

STATIC THEORY

- **Rotation** — spin on axis, ~24 hours (~1,675 kmph at equator) → **day & night**.
- **Revolution** — orbit round the Sun, **365.25 days** → **the year** (leap-year adjustment).
- **Axial tilt** = about **23.5° from the perpendicular to the orbital plane** (66.5° to the plane) → **seasons** and changing day length.
- □ Earth's geometric approximation = **oblate spheroid**; geoid = gravity-defined mean-sea-level surface.
- □ Rotation ≈ 24 h (day-night); Revolution = 365.25 days (year); tilt 23.5° (seasons).
- □ **15° longitude = 1 hour**; IST meridian = **82.5°E (UTC+5:30)**.
- □ International Date Line ≈ **180°** meridian.
- □ Only the **outer core is liquid** (S-wave proof); inner core is solid.
- □ Moho = crust/mantle; Gutenberg = mantle/outer-core.
- Explain how **seismic-wave (P/S) behaviour** maps the Earth's layered, differentiated interior — the
- The longitude–time relationship and the case for/against a **separate North-East India time zone**.
- □ India: **8°4'N–37°6'N** and **68°7'E–97°25'E**.
- □ **Tropic of Cancer (23.5°N)** passes through **8 Indian states**.
- □ IST meridian = **82°30'E** (near Mirzapur); **UTC+5:30**.
- □ ~2-hour sun-time gap between Arunachal and Gujarat, yet a single IST.

MAINS ANGLE

These are model content points; arrange them into paragraphs rather than reproducing a list.

STUDY LINK

India's Location & Extent -> 10-mark answer

HIGH

Solved 15-Mark Practice

GS-I Geography

INTRO & ORIGIN

Analyse India's Location & Extent with reference to its concepts, applications and limitations.

ANSWER ARCHITECTURE**1. Concept**

Define the exact scope and terminology.

V

2. Core analysis

Organise the source content into two or three dimensions.

V

3. Evidence

Use the chapter's India-centric examples or chronology.

V

4. Critical layer

Add traps, limitations, counter-view or implementation constraints.

V

5. Judgement

Conclude in direct response to the directive.

STATIC THEORY

- **Rotation** — spin on axis, ~24 hours (~1,675 kmph at equator) → **day & night**.
- **Revolution** — orbit round the Sun, **365.25 days** → **the year** (leap-year adjustment).
- **Axial tilt** = about **23.5° from the perpendicular to the orbital plane** (66.5° to the plane) → **seasons** and changing day length.
- ☐ Earth's geometric approximation = **oblate spheroid**; geoid = gravity-defined mean-sea-level surface.
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- ☐ **15° longitude = 1 hour**; IST meridian = **82.5°E (UTC+5:30)**.
- ☐ International Date Line ≈ **180°** meridian.
- ☐ Only the **outer core is liquid** (S-wave proof); inner core is solid.
- ☐ Moho = crust/mantle; Gutenberg = mantle/outer-core.
- Explain how **seismic-wave (P/S) behaviour** maps the Earth's layered, differentiated interior — the
- The longitude–time relationship and the case for/against a **separate North-East India time zone**.
- ☐ India: **8°4'N–37°6'N** and **68°7'E–97°25'E**.
- ☐ **Tropic of Cancer (23.5°N)** passes through **8 Indian states**.
- ☐ IST meridian = **82°30'E** (near Mirzapur); **UTC+5:30**.
- ☐ ~2-hour sun-time gap between Arunachal and Gujarat, yet a single IST.

MAINS ANGLE

These are model content points; arrange them into paragraphs rather than reproducing a list.

STUDY LINK

India's Location & Extent -> 15-mark answer

HIGH

Solved 20-Mark Practice

GS-I Geography

INTRO & ORIGIN

Critically examine India's Location & Extent with reference to its concepts, applications and limitations.

ANSWER ARCHITECTURE

1. Concept

Define the exact scope and terminology.

V

2. Core analysis

Organise the source content into two or three dimensions.

V

3. Evidence

Use the chapter's India-centric examples or chronology.

V

4. Critical layer

Add traps, limitations, counter-view or implementation constraints.

V

5. Judgement

Conclude in direct response to the directive.

STATIC THEORY

- **Rotation** — spin on axis, ~24 hours (~1,675 kmph at equator) → **day & night**.
- **Revolution** — orbit round the Sun, **365.25 days** → **the year** (leap-year adjustment).
- **Axial tilt** = about **23.5° from the perpendicular to the orbital plane** (66.5° to the plane) → **seasons** and changing day length.
- □ Earth's geometric approximation = **oblate spheroid**; geoid = gravity-defined mean-sea-level surface.
- □ Rotation ≈ 24 h (day-night); Revolution = 365.25 days (year); tilt 23.5° (seasons).
- □ **15° longitude = 1 hour**; IST meridian = **82.5°E (UTC+5:30)**.
- □ International Date Line ≈ **180°** meridian.
- □ Only the **outer core is liquid** (S-wave proof); inner core is solid.
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MAINS ANGLE

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STUDY LINK

India's Location & Extent -> 20-mark answer

HIGH**Final Consolidated Register Notes****GS-I Geography****INTRO & ORIGIN**

Final revision is placed after teaching and solved practice.

TOPIC RECALL MAP**India's Location & Extent**

V

Definition Rotation — spin on axis, ~24 hours (~1,675 kmph at equator) → day & night .	Mechanism Revolution — orbit round the Sun, 365.25 days → the year (leap-year adjustment).
Application Axial tilt = about 23.5° from the perpendicular to the orbital plane (66.5° to the plane) → seasons and changing day length.	Evaluation ☐ Earth's geometric approximation = oblate spheroid ; geoid = gravity-defined mean-sea-level surface.

STATIC THEORY

- **Rotation** — spin on axis, ~24 hours (~1,675 kmph at equator) → **day & night**.
- **Revolution** — orbit round the Sun, **365.25 days** → **the year** (leap-year adjustment).
- **Axial tilt** = about **23.5°** from the perpendicular to the orbital plane (66.5° to the plane) → **seasons** and changing day length.
- ☐ Earth's geometric approximation = **oblate spheroid**; geoid = gravity-defined mean-sea-level surface.
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- ☐ **15° longitude = 1 hour**; IST meridian = **82.5°E (UTC+5:30)**.
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- ☐ Only the **outer core is liquid** (S-wave proof); inner core is solid.
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- ☐ India: **8°4'N–37°6'N** and **68°7'E–97°25'E**.

MEMORY HOOK

Meaning -> mechanism -> evidence -> limitation -> judgement

MUST-KNOW FACTS

1. ☐ **Tropic of Cancer (23.5°N)** passes through **8 Indian states**.
2. ☐ IST meridian = **82°30'E** (near Mirzapur); **UTC+5:30**.
3. ☐ ~2-hour sun-time gap between Arunachal and Gujarat, yet a single IST.
4. ☐ India's officially remeasured coastline is **11,098.81 km**. The legacy
5. Use India's longitudinal extent to argue **for/against a separate North-East time zone** — energy and
6. How India's latitudinal spread (tropical to sub-tropical) shapes its climatic diversity.

UPSC TRAPS

WRONG: Earth is a perfect sphere

CORRECT: oblate spheroid/geoid.

WRONG: Rotation causes seasons

CORRECT: rotation causes day/night; **tilt + revolution** cause seasons.

WRONG: Latitude fixes time

CORRECT: longitude fixes time.

WRONG: The whole core is molten

CORRECT: only the outer core is liquid.

MAINS ANGLE

Use India's Location & Extent as an argument, not as an unstructured list of facts.

STUDY LINK

Geography -> India's Location & Extent

End of Register Notes - UPSC Agent / Copilot CLI